

Unit 4

Derivative Applications – Optimization

Position, Velocity, and Acceleration

The **position** function $s(t)$ is where an object is at a particular time.

- Distance an object has moved from a reference point (starting point) over a period of time t .
- In Physics, we call this the **displacement**.

The **velocity** function $v(t)$ is how quickly the position is changing.

The **acceleration** function $a(t)$ is how quickly the velocity is changing.

	Displacement $s(t)$	Velocity $v(t)$	Acceleration $a(t)$
Relationship	$s(t)$	$v(t) = s'(t)$	$a(t) = s''(t)$
Units	m	$\frac{m}{s}$	$\frac{m}{s^2}$

When setting up word problems, direction is important

- From the reference point, moving:

Up is +ve	Right is +ve
Down is -ve	Left is -ve

	-	+
$s(t)$	You are to the <u>left</u> of the <u>R.P</u> or/ <u>below</u> the <u>R.P</u>	You are to the <u>right</u> of the <u>R.P</u> or/ <u>above</u> the <u>R.P</u>
$v(t)$	You are <u>moving</u> to the <u>left</u> or/ <u>moving</u> <u>lower</u>	You are <u>moving</u> to the <u>right</u> or/ <u>moving</u> <u>higher</u>
$a(t)$	You are being <u>pulled</u> to the <u>left</u> or/ <u>pulled</u> <u>down</u>	You are being <u>pulled</u> to the <u>right</u> or/ <u>pulled</u> <u>up</u>

If velocity and acceleration are working in the same direction, the object is **speeding up**

- Speeding up: $v(t) * a(t) > 0$

If velocity and acceleration are working in the opposite direction, the object is **slowing down**

- Slowing down: $v(t) * a(t) < 0$

	$a(t) < 0$	$a(t) > 0$
$v(t) < 0$	Moving left and being pulled left • Speed <u>increase</u> to the <u>left</u>	Moving left and being pulled right • Speed <u>decrease</u> to the <u>left</u>
$v(t) > 0$	Moving right and being pulled left • Speed <u>decrease</u> to the <u>right</u>	Moving right and being pulled right • Speed <u>increase</u> to the <u>right</u>

Ex 1) A little girl is very upset because she cannot open her lunch box. In frustration, she jumps to the top of the school (9 m high) and throws her lunch box upwards with a velocity of 12 m/s. She watches her lunch box crash to the pavement below. The height of her lunch box in metres is given by $h(t) = -4.9t^2 + 12t + 9$, where t is in seconds.

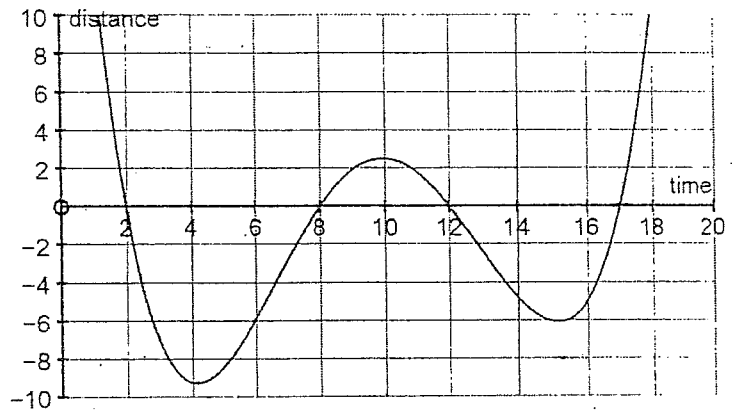
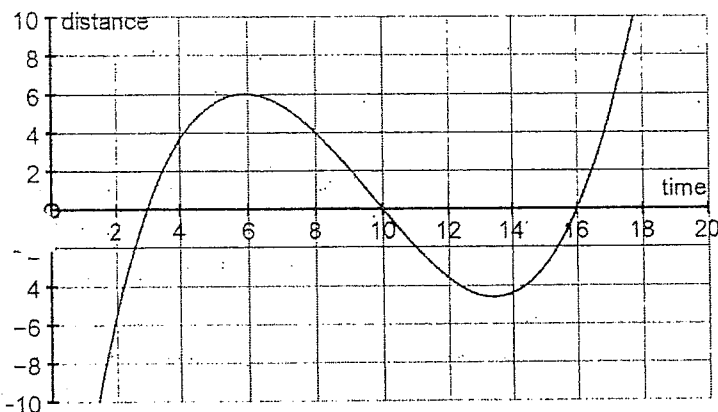
a) How long does it take for the lunch box to hit the ground? [3.05 seconds]

b) How high does her lunch box reach? [16.3 m]

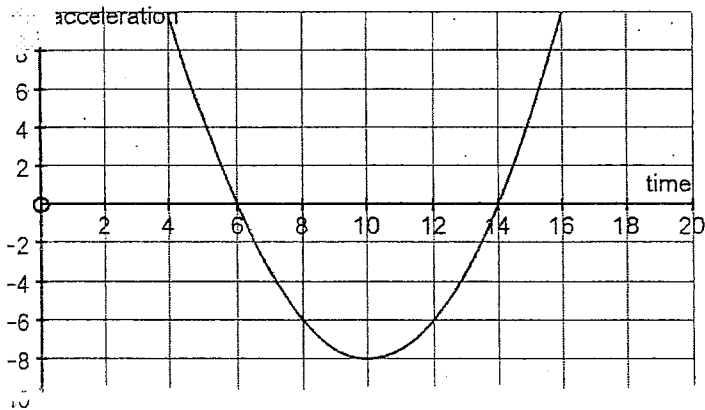
c) If her lunch box can withstand an impact velocity of 20 m/s, will her lunch box break open so she can eat her lunch? [no]

d) What is the acceleration of her lunch box?

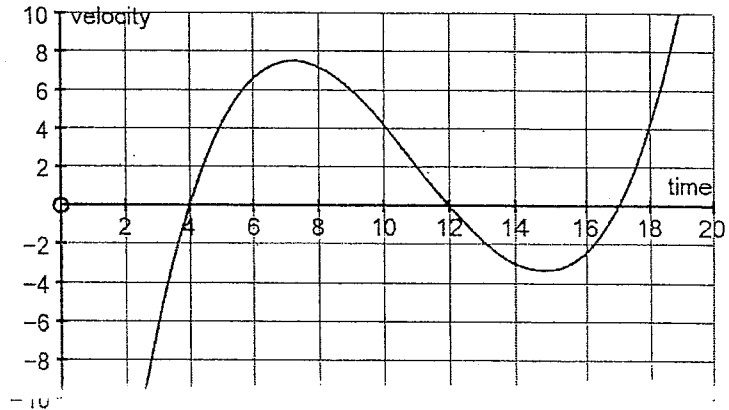
Ex) For the given graphs, sketch the velocity and acceleration functions



Ex) For the given graph, sketch the distance and velocity functions



Ex) For the given graph, sketch the distance and acceleration functions



Ex) A particle moves according to $s(t) = 2t^3 - 21t^2 + 60t - 12$ where s is in metres and t is in seconds.

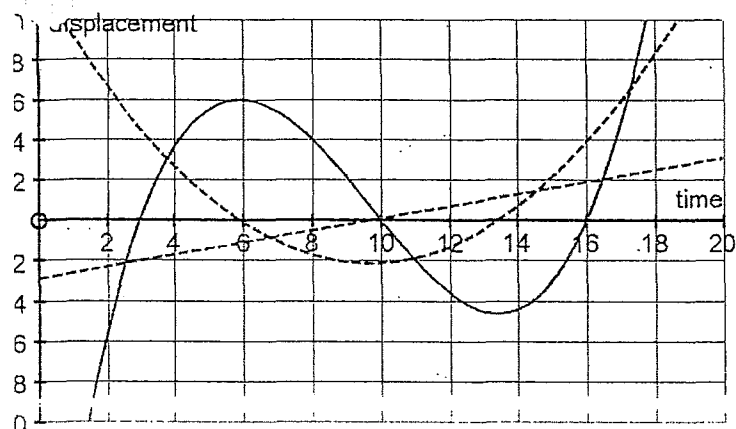
a) When does the particle change direction? $[t = 2, 5]$

b) How quickly is the particle moving after 3 seconds? $\left[-12 \frac{m}{s}\right]$

c) Is the particle speeding up or slowing down at 4 seconds? $[slowing\ down]$

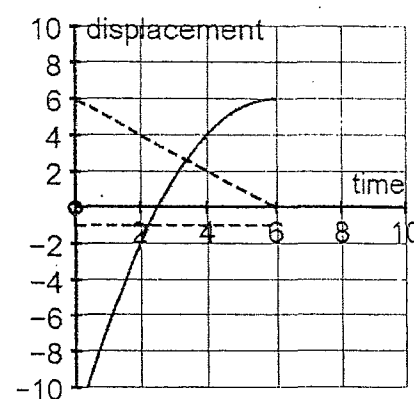
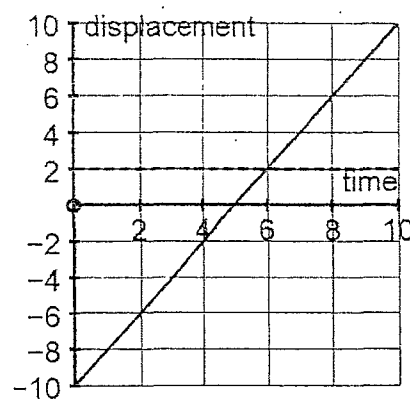
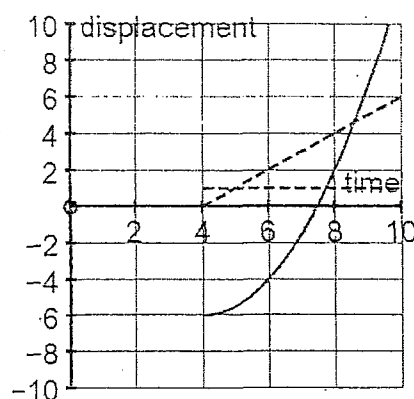
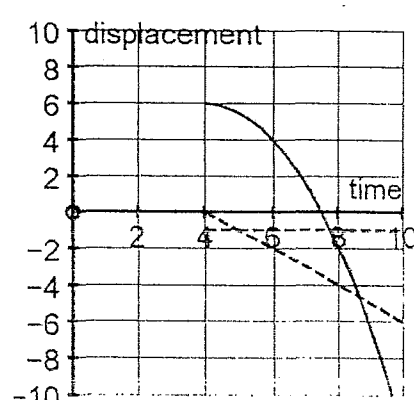
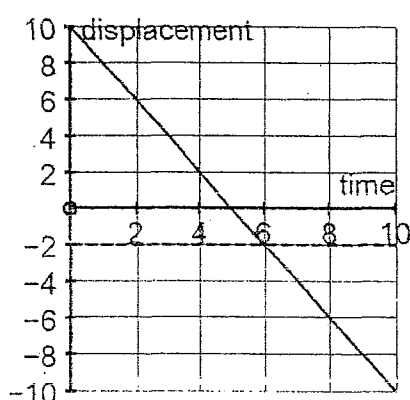
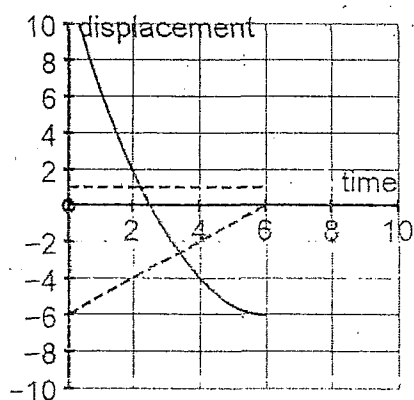
d) Sketch the motion of the particle.

Ex) Given the following displacement/velocity/acceleration time graph, state the intervals when the object is speeding up and slowing down.



Ex) In each scenario, state

- If the function is increasing or decreasing and the sign of the velocity
- The concavity and the sign of the acceleration
- Whether the object is speeding up or slowing down.



Warm-Up

1. The position function of an object moving along a straight line is $s(t) = t^3 - 24t^2 + 144t + 15$ where t is in seconds and s is in metres. Assume that the starting position of 15 m is the reference point.

a) When does the object change direction?

b) Is the object speeding up or slowing down at $t = 9s$

c) Determine the displacement of the object after $t = 5s$ and sketch the motion of the particle.

The Second Derivative Test

If you are given a function, you can use the second derivative to determine if a function has a local maximum or a local minimum at a critical number c , using the second derivative.

Steps:

1. Find the critical numbers of $f'(x)$ ie) $f'(c) = 0$ or $f'(c)$ is DNE.
2. Sub the critical number into $f''(x)$
 - If $f''(c) > 0$, then local min at c
 - If $f''(c) < 0$, then local max at c
 - If $f''(c) = 0$, you cannot conclude anything. You must use an interval chart for f'

Number Problems

1. The sum of two positive numbers is 12. If the product of one number by the cube of the other is to be a maximum, find the numbers. (9, 3)
2. The sum of a positive number and its reciprocal is a minimum. What is the number? (1)
3. Find two numbers whose difference is 150 and whose product is a minimum. (-75, 75)
4. Find two positive numbers with product 200 such that the sum of one number and twice the second number is as small as possible. (10, 20)
5. The sum of two numbers is k . Show that the sum of their squares is at least $\frac{1}{2}k^2$.

Perimeter and Area Problems

1. Three sides of a rectangular playing field are to be fenced with 400m of fencing. Find the dimensions so that the area of the field will be a maximum. (200m × 100m)
2. What is the maximum possible area of an isosceles triangle if the two equal sides are each 15 cm long? (112.5 cm²)
3. Two equal rectangular playing fields, having one side in common, are to be enclosed (and divided) by 600m of fencing. Find the largest possible area for each field. (100m × 75m = 7 500 m²)
4. A rectangular field along a straight river is to be divided into three smaller fields by one fence parallel to the river and four fences perpendicular to the river. Find the maximum area that can be enclosed if 1600m of fencing is available. (200m × 800m)
5. An experimental farm has 3600m of fencing with which to enclose and sub-divide a rectangular field into 4 equal plots of land.
 - (a) What is the largest area that can be enclosed? (360m × 225m)
 - (b) What is the largest area that can be enclosed if each side of each plot is required to be at least 250m long? (320 000 m²)
6. A supermarket is designed to have a rectangular floor area of 3750 m², with three walls made of cement blocks and one wall made of glass. In order to conform to the building code, the length of the glass wall must not exceed 60m but must not be 30m. The cost per linear metre of constructing a glass wall is twice that of constructing a cement wall. Find the dimensions of the floor area that will minimize the cost of building the walls. (50m glass part × 75m)

Maximum and Minimum Problems: AREA and VOLUME

1. A farmer has 54m of fencing with which to build two animal pens with a common side. One pen is rectangular, and the other is a square. If the areas of the pens is to be maximized, what are their dimensions?
(9m $\times$ 9m, 9m $\times$ 4.5m)
2. A box with an open top is to be made from a square piece of cardboard of side length 100 cm, by cutting a square piece from each corner and then folding up the sides. Find the dimensions of the box of largest volume.
 $\left(\frac{200}{3}\text{m} \times \frac{200}{3}\text{m} \times \frac{50}{3}\text{m}\right)$
3. Find the dimensions of the base of a rectangular box of greatest volume that can be constructed from 200 cm² of cardboard if the base is to be three times as long as it is wide.
(10 cm $\times$ 3 $\frac{1}{3}$ m)
4. A manufacturer wishes to produce cylindrical fruit juice cans with a capacity of 250 mL. What dimensions will minimize the amount of material required for a can? (1 mL = 1 cm³)
r = 3.414 cm, h = 6.828 cm)
5. Find the largest open-topped rectangular box that has a square base and a total surface area of 3600 cm². What is its volume?
(34.6cm $\times$ 34.6cm $\times$ 17.3 cm; 20 760 cm³)
- omit { 6. A cone-shaped paper drinking cup is to hold 36 cm³ of water. Find the height and radius of the cup that will require the least amount of paper to make. The lateral surface area of a cone is $A = \pi rs$, where r is the radius and s is the slant height of the cone.
(r = 2.9 cm, h = 4.1 cm)
7. Determine the dimensions of the cylinder of maximum volume that can be inscribed in a sphere of radius 10 cm.
(r = 8.1 cm, h = 11.55 cm)

BEST TIME PROBLEMS

1. A man in a boat is 3 km offshore and wishes to go to a point on the shore that is 5 km from the point directly opposite his present position. The man can walk at 4 km/h and row at 2 km/h. At what point on the shore should he land so that he can reach his destination in the shortest possible time?
(3.27 km)
2. A sailor in a boat 8 km off a straight coastline wants to reach a point on shore 10 km from the point directly opposite her present position in the shortest possible time. Towards what point on the shore should she steer and how long does it take her to reach her destination if
 - (a) she can row at 4 km/h and run at 6 km/h? (2.84 km; 3.15h)
 - (b) she can row at 4 km/h and walk at 5 km/h?
(x < 10 km ; row all the way)
3. A pair of scuba divers wish to dive on a wreck that lies 0.4 km off the shore from a point 0.8 km from their present position. If they can walk carrying their gear at 5 km/h and swim at 3 km/h, what course should they follow to reach the wreck in the minimum time? (walk 0.5 km, swim the rest)
4. A fisherman wishes to cross a stream 200m wide in his row boat and then to reach a place on the far bank of the river 500 m away from the point directly opposite his launching point. If he can row at 3 m/s and walk at 5 m/s, find his shortest time to reach the desired place. (153.3 s = 2.56 min.)
5. Two isolated farms are situated 12 km apart on a straight country road that runs parallel to the main highway 20 km away. The power company decides to run a wire from the highway to a junction box, and from there, wires of equal lengths to the two houses. Where should the junction box be placed to minimize the length of wire needed?

REVENUE PROBLEMS

1. A car rental agency has 200 cars. The owner finds that at a price of \$36 per day, he can rent all the cars. For each \$2 increase in price, the demand is less and 5 fewer cars are rented. What price will maximize the total revenue? (\$58/day)
2. The current ticket price at a local theatre is \$4, and the theatre attracts an average of 250 customers per show. Every 20 cent increase in ticket price reduces to average attendance by 10 customers, while every 20 cent decrease results in 10 extra customers. Find the ticket price that will maximize the revenue. (\$4.50)
3. A variety store can sell 500 yo-yos for \$1 apiece. For each penny that the store lowers the price, it can sell 20 more yo-yos. For what price should it sell the yo-yos to maximize its profit? (62 cents)
4. A retailer of electrical appliances can sell 200 refrigerators at \$250 each. For each reduction of \$10 in price, 10 more refrigerators are sold (Reductions of less than \$10 per unit have no effect on sales). What selling price would produce the maximum sales per month, and how many refrigerators would be sold at this price? (230 at \$220 or 220 at \$230)
5. A real estate firm owns 250 apartments that can be rented out at \$460 per month each. For each \$5 per month increase in rent, there are two vacancies created that can not be filled. What should the monthly rent be to maximize revenue? What is the maximum revenue? (\$542.50; \$117 722.60)
6. A rectangular building is to be constructed with a curtain wall that will be parallel to the front and will divide the building into a sales area of 1500 m^2 . The outside walls at both sides and across the back will cost \$100 per linear metre. The curtain wall across the store will cost \$45 per linear metre. The front wall (mainly glass) will cost \$345 per linear metre. Calculate the dimensions of the rectangular building that will give the required floor areas at the lowest cost for the walls. The front wall should be at least 20 m wide. $(70\text{m} \times \frac{200}{7}\text{m})$